### 1.2 Drive System Choices

#### 1.2.1 Motor Selection

The following table shows the comparison between the LEGO EV3 Medium Motor and EV3 Large Motor based on four main criteria.

| Criteria | EV3 Medium Motor | EV3 Large Motor |
|---|---|---|
| **Motor Photo** |  |  |
| **Torque (Running Torque)** | 6.64 N·cm | 17.3 N·cm |
| **Speed (No-load)** | 165 RPM | 105 RPM |
| **Part Number** | 45503 | 45502 |
| **Key Feature** | Compact and lightweight; higher rotational speed | Higher torque; suitable for heavy loads |

- We selected the **EV3 Medium Servo Motor** because our robot weighs approximately 800 g and does not require the high torque provided by the EV3 Large Servo Motor.
- With a maximum speed of 165 RPM, the Medium Motor offers higher rotational speed compared to the Large Motor (105 RPM). In our actual operation, we typically use only 65% of the motor speed, which is sufficient to move our robot effectively.
- Therefore, the Medium Motor provides an ideal balance of speed, torque, and compactness without requiring the additional torque of a larger motor.

**Testing Performance for Motor Selection**

| Parameters | EV3 Medium Motor | EV3 Large Motor |
|---|---|---|
| **Motor Speed** | 100% | 100% |
| **Total Testing Rounds** | 10 | 10 |
| **Successful Rounds** | **9** | **2** |
| **Failed Rounds** | 1 | 8 |
| **Success Rate** | 90% | 20% |
| **Average Completion Time** | 25.67s | 18s |

**Engineering Decision**:

- The EV3 Medium Motor was selected instead of the EV3 Large Motor based on its higher reliability during the Open Challenge testing at 100% speed.
- The Medium Motor achieved a 100% success rate (6/6 trials) with an average completion time of 25.67 seconds, while the Large Motor achieved only a 20% success rate (2/10 trials) despite being faster, with an average completion time of 18 seconds.
- Although the Large Motor offered better speed, the Medium Motor provided more consistent performance.
- Therefore, reliability and successful navigation were prioritised over maximum speed.

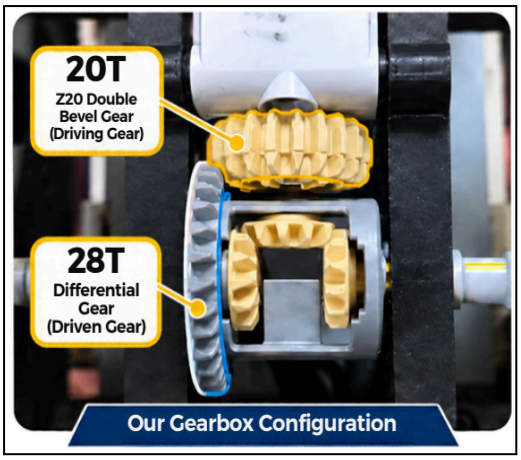
### 1.2.2 Gear Combination Selection

To further minimize the drivetrain size, we designed a compact gearbox using:

| | | |
|---|---|---|
|  |  |  |
| 1 piece Z20 Double Bevel Gear (20 Teeth) – Part 18575 | 1 piece Differential Gear (28 Teeth) – Part 62821 | 3 piece Conical Wheels (12 Teeth) – Part 6589 |

**Gear Ratio Calculation**

| |
|---|
| DrivenGearTeeth = 28T |
| DrivingGearTeeth = 20T |
| Motor Speed = 165RPM |
| Torque = 6.64 N.CM |




$$GearRatio = \frac{DrivenGearTeeth}{DrivingGearTeeth} = \frac{28T}{20T} = 1.4:1$$

$$OutputSpeed = \frac{InputSpeed}{GearRatio} = \frac{165}{1.4} = 117.86RPM$$

$$OutputTorque = Motor\ Torque \times Gear\ Ratio = 6.64 \times 1.4 = 9.296\ \text{N·cm}$$

- We selected a 1.4:1 gear reduction to balance motor speed and output torque. The 20T Z20 Double Bevel Gear transfers power from the EV3 Medium Motor to the 28T Differential Gear.
- This reduction decreases rotational speed while increasing the theoretical output torque.
- The differential gearbox, which contains three internal 12T bevel gears, distributes power to the left and right drive wheels while allowing them to rotate at different speeds during turns.
- This helps the robot navigate corners smoothly while maintaining stable movement.

### 1.2.3 Drive System Choices Reasoning（Ideal Force & Thrust-to-Weight Ratio)

**Selected Drive System Parameters**

| Parameter | Selected Value/Items |
|---|---|
| **Motor** | EV3 Medium Motor |
| **Gear Ratio** | 1.4:1 |
| **Motor Torque** | 6.64 N·cm |
| **Motor Speed** | 165 RPM |
| **Output Torque,$T_{out}$** | 9.296 N·cm |
| **Output Speed** | 117.86 RPM |
| **Wheel** | LEGO 32019 + 86652 |
| **Wheel Radius,r** | 31.2 mm |
| **Robot Mass,m** | 800g |
| **Gravity (g)** | 9.81 m/s² |

| Driving Force Calculation (F) | Thrust-to-Weight Ratio (TWR) |
|---|---|
| $F = \frac{T_{out}}{r}$<br>$= \frac{0.09296}{0.0312}$<br>$= 2.98\text{N}$ | $TWR = \frac{F}{m \times g}$<br>$= \frac{2.98}{0.8 \times 9.81}$<br>$= 0.38:1$ |

- The EV3 Medium Motor with a 1.4:1 gear reduction was selected because it provides a suitable balance between driving force and output speed.
- The calculated driving force of 2.98 N and TWR of 0.38:1 are considered suitable for the 0.8 kg robot, while maintaining a compact drivetrain design.